Parth Nitesh Chaudhari

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Education

International Institute of Information Technology, Hyderabad — Hyderabad, India
B.Tech in Computer Science and Engineering — *July 2024 – May 2028*

Experience

Undergraduate Research Assistant — April 2026 – Present
Data Science and Analytics Centre, IIIT-H (Prof. Kamalakar Karlapalem) — *Python, scikit-learn, OpenCV*

- Modelled how defensive shape drives chance creation using tracking data from **17 pro matches** in two leagues: **8,087** phases, 339 shots, 9 shape features every 0.5 s, logistic models adjusted for ball position and 1,000-resample bootstraps over games.
- Counter-attacks were the most shape-sensitive: with few defenders behind the ball, **51% vs 27%** ended in a shot (A-League) and chances were **7.4**× higher in xG (Bundesliga). A shape-only model reached AUC 0.47–0.63 on unseen games: shape shifts the odds but does not decide single attacks.
- Labelled **24,374** passing options as clear, contested or blocked from player geometry; on 5,803 held-out passes completion fell from **86% to 62%** with 2+ defenders in the lane, and counting defenders beat a yes/no rule (AUC 0.61 vs 0.53).
- Built an OpenCV pipeline with logistic expected-pass and xG models on broadcast clips; ran it on **36,750 unseen frames** on the HPC cluster, then showed its screen-space features tracked the camera, not the game, and moved the study to pitch coordinates.

Software Engineering Team Project — January 2026 – May 2026
Cosmic 3D, Maker's Lab (client project, DASS course) — *TypeScript, React, MQTT, ESP32*

- 5-person team building remote control and live monitoring for ESP32 3D printers; owned the MQTT layer (Mosquitto over TCP and WebSockets) that streams printer telemetry to the React dashboard.

Projects

LLM Confidence Under Pressure | *PyTorch, Hugging Face, Slurm, W&B* — September 2026 – October 2026

- Ran Qwen2.5-7B and Llama-3.1-8B (4-bit) through six turns of criticism, praise or a neutral re-ask on **450 claims** of known truth (MATH-500, real analysis), logging verdicts, token-probability probes and stated confidence every turn.
- Under sustained criticism Llama kept only **14.7%** of correct MATH answers vs **68.6%** in the control (Qwen: 76.3% vs 93.9%), and its true/false discrimination fell from 0.46 to **0.11**. Harsher tone mattered only when it also called the answer wrong.
- When Qwen gave in, its own working still reached the right answer **73%** of the time; linear probes on hidden states (5-fold, grouped by claim pair) tested whether Llama kept a trace of the correct answer.
- Fully deterministic: pinned model revisions, greedy decoding, fixed seeds; every report number rebuilds from the logged runs without a GPU.

Event Management Platform | *React, Express, MongoDB, Socket.IO* — February 2026

- Built solo and deployed a **7,800-line** full-stack platform: **52 REST endpoints**, JWT role-based portals, real-time Socket.IO chat, QR e-tickets by email and a custom form builder.

CLI-Docs: Distributed Document Store | *C, pthreads, TCP sockets* — October 2025 – November 2025

- **4,700-line** C system (2-person team): name server, multi-threaded storage servers with per-sentence write locks, O(1) hash lookups, persisted metadata, undo and streaming.

C-Shell and Network Protocol Suite | *C, POSIX, UDP, libpcap* — September 2025 – October 2025

- POSIX shell with pipelines, redirection and job control; TCP-like sliding-window transport over UDP; libpcap packet sniffer with BPF filters.

Technical Skills

Languages: Python, C, C++, SQL, TypeScript, JavaScript
ML / LLMs: PyTorch, Hugging Face Transformers, bitsandbytes, scikit-learn, OpenCV
Methods: LLM evaluation, hidden-state probing, logistic regression, bootstrap confidence intervals, cross-validation
Tools: Slurm, Weights & Biases, Git, Linux, pandas, NumPy, Matplotlib